PROPOSED EVIDENCE-BASED FRAMEWORK FOR STANDARDIZED NEOANTIGEN IDENTIFICATION

An adaptable, platform-independent workflow derived from clinical translation evidence and scoping review consensus

PHASE I: UPSTREAM IMMUNOGENOMICS (WET LAB & VARIANT CALLING)

PHASE II: DOWNSTREAM TRANSLATION (SELECTION & CLINICAL TRANSITION)

01 | SPECIMEN & SEQUENCING

- **Tumor-Normal Matched Design**
  Mandatory comparison to filter germline variants.
- **Exome Sequencing (WES)**
  High depth (>100x tumor, >50x normal) standard.
- **Complementary RNA-Seq**
  Required to verify tumor expression of variants.
- **Illumina Platform Prevalence**
  Establishes standard baseline for sequencing.

Ensemble Variant Calling

02 | SOMATIC VARIANT DISCOVERY

- **Multi-Caller Ensemble Approach**
  Combine MuTect2, Strelka2, and VarScan2.
- **Variant Annotation Databases**
  Use Ensembl VEP or ANNOVAR platforms.
- **Variant Quality Filtering**
  Rigorous read support and alignment checks.
- **Expression Level Filtering**
  TPM > 10 threshold to ensure transcription.

HLA Allele Restriction

03 | HLA & EPITOPE PREDICTION

- **High-Resolution HLA Genotyping**
  Utilize OptiType (Class I) and PolySolver.
- **MHC Class I Standard Prediction**
  Adopt NetMHCpan v4.0/4.1 family algorithms.
- **Mandatory Class II Prediction Integration**
  Incorporate HLA-II to capture helper CD4+ T-cells.
- **Pan-specific Neural Networks**
  Leverage mass spec-trained affinity models.

Candidate Selection

Prediction

Prioritization

04 | CANDIDATE PRIORITIZATION

- **Multi-Parameter Ranking Paradigm**
  Discard single-affinity (IC50) reliance.
- **Key Prioritization Criteria:**
  1. Binding Affinity (IC50 < 500 nM)
  2. Percentile Rank (< 2.0% standard)
  3. RNA expression abundance (TPM)
  4. Mutation clonality (VAF, subclonal check)
  5. TAP processing, stability, & presentation

05 | EXPERIMENTAL VALIDATION

- **Functional Immunogenicity Verification**
  Validate de novo T-cell activation post-vaccine.
- **Core Immunological Assays**
  IFN-g ELISpot and Intracellular Cytokine Staining.
- **Peptide-MHC Multimer Assays**
  Direct staining to detect antigen-specific T-cells.
- **TCR Repertoire Sequencing**
  Track clone expansion and diversity (MiXCR).

Immunogenicity Assay

06 | REPORTING & CLINICAL TRANS

- **Methodological Transparency (FAIR)**
  Report versions, parameters, and reference genomes.
- **Delivery Platforms in Clinical Use**
  Synthetic long peptide (SLP), RNA/saRNA, DC.
- **Safety-focused Early-Phase Layout**
  Target Phase I/Ib designs with ICI combinations.
- **Standardized Clinical Reporting**
  Pre-enrollment screening & HLA diversity reporting.

Translational Application